

CSS Pseudo-Classes & Pseudo-Elements

Clear, Simple Notes with Syntax and Examples

1. What are Pseudo-Classes and Pseudo-Elements?

In CSS, pseudo-classes and pseudo-elements allow us to style HTML elements based on their state, position, or specific sub-parts without changing the HTML code.

Key Difference

Pseudo-Class (`:`) → Styles an existing element based on a **STATE** or **POSITION**.

Pseudo-Element (`::`) → Styles a **SPECIFIC PART** or inserts dynamic content.

2. CSS Pseudo-Classes (`:`)

A pseudo-class is used to define a special state of an element.

Syntax

```
selector:pseudo-class {  
  property: value;  
}
```

Common Types & Examples

A. User Action State

Changes how an element looks when a user interacts with it.

- `:hover` — When the user moves the mouse cursor over an element.
- `:focus` — When an element receives focus (clicking or tabbing into an input).
- `:active` — When an element is being clicked.

```

/* Mouse hover effect on a button */
button:hover {
    background-color: blue;
    color: white;
}

/* Input box when active/focused */
input:focus {
    border: 2px solid green;
}

```

B. Link States

Used to style hyperlinks depending on whether they have been visited.

Unvisited (:link) → Visited (:visited) → Hover (:hover) →
Active (:active)

```

a:link { color: blue; }
a:visited { color: purple; }
a:hover { color: red; }
a:active { color: yellow; }

```

C. Structural Position

Selects elements based on where they are located inside their parent element.

- :first-child — Selects the very first child element.
- :last-child — Selects the last child element.
- :nth-child(n) — Selects elements using a number or pattern (like even/odd).

```

/* Style the first item in a list */
li:first-child {
    font-weight: bold;
}

/* Alternate background colors for table rows */
tr:nth-child(even) {
    background-color: #f2f2f2;
}

```

3. CSS Pseudo-Elements (::)

A pseudo-element is used to style specific parts of an element, or insert content before/after it.

Syntax

```
selector::pseudo-element {  
    property: value;  
}
```

Common Types & Examples

A. Generated Content (::before and ::after)

Inserts content before or after the text of an element. The `content` property is required.

[`::before`] + [Element Content] + [`::after`]

```
/* Add a quote symbol before blockquotes */  
blockquote::before {  
    content: "\" ";  
    color: gray;  
}  
  
/* Add an indicator after a special class */  
.new-badge::after {  
    content: " (NEW)";  
    color: red;  
}
```

B. Text Parts (::first-letter and ::first-line)

Targets the very first letter or line of a block of text.

```
/* Drop cap effect for the first letter */  
p::first-letter {  
    font-size: 200%;  
    color: blue;  
}  
  
/* Style the first line of a paragraph */  
p::first-line {  
    font-weight: bold;  
}
```

C. User Selection (`::selection`)

Styles text that is highlighted/selected by the user.

```
::selection {  
    background-color: yellow;  
    color: black;  
}
```

4. Quick Summary Matrix

Feature	Pseudo-Class	Pseudo-Element
Symbol	Single colon (<code>:</code>)	Double colon (<code>::</code>)
Purpose	Styles an element based on state or position	Styles a specific sub-part or inserts content
Examples	<code>:hover</code> , <code>:focus</code> , <code>:nth-child()</code>	<code>::before</code> , <code>::after</code> , <code>::first-letter</code>